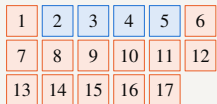


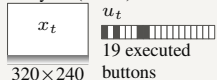
(a) Data

Arnold agent vs. 8 bots,
150 s episodes on the 17
arenas of one WAD



■ 4 training maps, 500 ep.
■ 13 unseen arenas, 24 ep.

every tic (35 Hz):



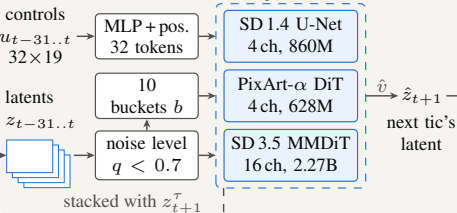
frozen VAE encoder E
(padded to 320×256)

latents $z_t \in \mathbb{R}^{C \times 32 \times 40}$
 $C=4$ (SD 1.x)
or 16 (SD 3.5)

(b) One recipe, three pretrained backbones

v -prediction, 200k updates of batch 32,
AdamW 5×10^{-5} , fp32 EMA 0.9999

denoiser f_θ : one of three



(c) After pretraining

Arena adaptation. Frozen backbone, rank-16 LoRA on attention; control MLP, positions, input proj., bucket table in full (4.2M, 0.49%); 8 episodes, 4k or 8k updates

Decoder fine-tune. E frozen; decoder D fit to training-map frames from their true latents $E(x)$ with MSE + 0.1 LPIPS

(d) Evaluation

one tic ahead from 32 true tics,
clean context ($b=0$), 10-step DDIM; 256 held-out windows per unseen arena, 512 on the training maps; scene rows 0–207, no HUD

A advantage over copy-last
 M LPIPS margin to persistence
 G gap to the reconstruction
 S_0 zero-shot latent skill
(M, G against the raw frame x)

